

flspcell_typ_1p50V_25C Library

Cell Groups
DFRBPQ

DFRBPQ



*flspcell_typ_1p50V_25C Cell Library: Process
flspcell_typ_1p50V_25C, Voltage 1.50, Temp 25.00*

Truth Table

INPUT			OUTPUT
CLK	D	RESET_B	Q
R	0	1	0
R	1	1	1
x	x	0	0
x	x	1	IQ

Footprint

Cell Name	Area
sg13g2_dfrbpq1arx_1	48.98880

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	CLK	D	RESET_B	Q
sg13g2_dfrbpq1arx_1	0.02223	0.01528	0.02768	0.30000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_dfrbpq1arx_1	7062.82000	8058.81000	8964.81000

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RR)	0.01860	0.00100	0.25239	0.32940	0.06480	0.44058	2.50740	0.30000	1.02711

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK->Q (RF)	0.01860	0.00100	0.27357	0.32940	0.06480	0.45569	2.50740	0.30000	0.93065
	RESET_B->Q (FF)	0.01860	0.00100	0.32322	0.32940	0.06480	0.54578	2.50740	0.30000	1.17507

Constraint Information

Constraints(ns) for CLK rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.07980	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for CLK falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	CLK ()	0.01860	0.00000	0.07980	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.03668	1.26300	1.26300	-0.12952	2.50740	2.50740	-0.16824
	setup	CLK (R)	0.01860	0.01860	0.31298	1.26300	1.26300	0.35618	2.50740	2.50740	0.36009

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	hold	CLK (R)	0.01860	0.01860	-0.01956	1.26300	1.26300	-0.14031	2.50740	2.50740	-0.22727
	setup	CLK (R)	0.01860	0.01860	0.33988	1.26300	1.26300	0.43983	2.50740	2.50740	0.51652

Constraints(ns) for RESET_B rising :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	recovery	CLK (R)	0.01860	0.01860	0.07336	1.26300	1.26300	0.19698	2.50740	2.50740	0.29220
	removal	CLK (R)	0.01860	0.01860	-0.06847	1.26300	1.26300	-0.19158	2.50740	2.50740	-0.28925

Constraints(ns) for RESET_B falling :

Cell Name	Timing Check	Ref Pin(trans)	Constraint(ns)								
			Input Slew(ns)	Ref Slew(ns)	First	Input Slew(ns)	Ref Slew(ns)	Mid	Input Slew(ns)	Ref Slew(ns)	Last
sg13g2_dfrbpq1arx_1	min_pulse_width	RESET_B ()	0.01860	0.00000	0.07660	1.26300	0.00000	2.08496	2.50740	0.00000	4.13818

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.22531	0.32940	0.06480	0.23566	2.50740	0.30000	0.38563

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.00100	0.24886	0.32940	0.06480	0.26419	2.50740	0.30000	0.40882
	RESET_B	0.01860	0.00100	0.23248	0.32940	0.06480	0.23854	2.50740	0.30000	0.29819

Passive power(pJ) for CLK rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.05298	0.32940	0.06496	2.50740	0.20474

Passive power(pJ) for CLK falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.09799	0.32940	0.11203	2.50740	0.26118

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.03698	0.32940	0.04870	2.50740	0.18937
	(D * !RESET_B * * !Q)	0.01860	0.05298	0.32940	0.06496	2.50740	0.20474
	(!D * RESET_B * !Q)	0.01860	0.05501	0.32940	0.06717	2.50740	0.20723
	(!D * !RESET_B * * !Q)	0.01860	0.06359	0.32940	0.07557	2.50740	0.21531

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(D * RESET_B * Q)	0.01860	0.09307	0.32940	0.10715	2.50740	0.25642
	(D * RESET_B * !Q)	0.01860	0.09799	0.32940	0.11203	2.50740	0.26118
	(D * !RESET_B * * !Q)	0.01860	0.04673	0.32940	0.06040	2.50740	0.20506
	(!D * RESET_B * Q)	0.01860	0.28791	0.32940	0.29349	2.50740	0.43722
	(!D * RESET_B * !Q)	0.01860	0.04094	0.32940	0.05469	2.50740	0.19909
	(!D * !RESET_B * * !Q)	0.01860	0.03627	0.32940	0.04990	2.50740	0.19454

Passive power(pJ) for D rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.08285	0.32940	0.08658	2.50740	0.14095

Passive power(pJ) for D falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.08596	0.32940	0.09030	2.50740	0.14767

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.08285	0.32940	0.08658	2.50740	0.14095
	(!CLK * RESET_B)	0.01860	0.14199	0.32940	0.14579	2.50740	0.20282
	(!CLK * !RESET_B)	0.01860	0.08694	0.32940	0.08905	2.50740	0.12672

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	CLK	0.01860	0.08596	0.32940	0.09030	2.50740	0.14767
	(!CLK * RESET_B)	0.01860	0.11517	0.32940	0.11948	2.50740	0.18049
	(!CLK * !RESET_B)	0.01860	0.06676	0.32940	0.06929	2.50740	0.10924

Passive power(pJ) for RESET_B rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.01546	0.32940	0.01807	2.50740	0.06696

Passive power(pJ) for RESET_B falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	0.01860	0.04041	0.32940	0.04341	2.50740	0.12472

Passive power(pJ) for RESET_B rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.01546	0.32940	0.01807	2.50740	0.06696
	(CLK * !D * !Q)	0.01860	0.00658	0.32940	0.00662	2.50740	0.00666
	(!CLK * D * !Q)	0.01860	0.07086	0.32940	0.07368	2.50740	0.14911
	(!CLK * !D * !Q)	0.01860	0.01186	0.32940	0.01191	2.50740	0.01194

Passive power(pJ) for RESET_B falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_dfrbpq1arx_1	(CLK * D * !Q)	0.01860	0.27435	0.32940	0.28333	2.50740	0.42990
	(CLK * !D * !Q)	0.01860	-0.00658	0.32940	-0.00662	2.50740	-0.00666
	(!CLK * D * !Q)	0.01860	0.04041	0.32940	0.04341	2.50740	0.12472
	(!CLK * !D * !Q)	0.01860	-0.01186	0.32940	-0.01191	2.50740	-0.01194